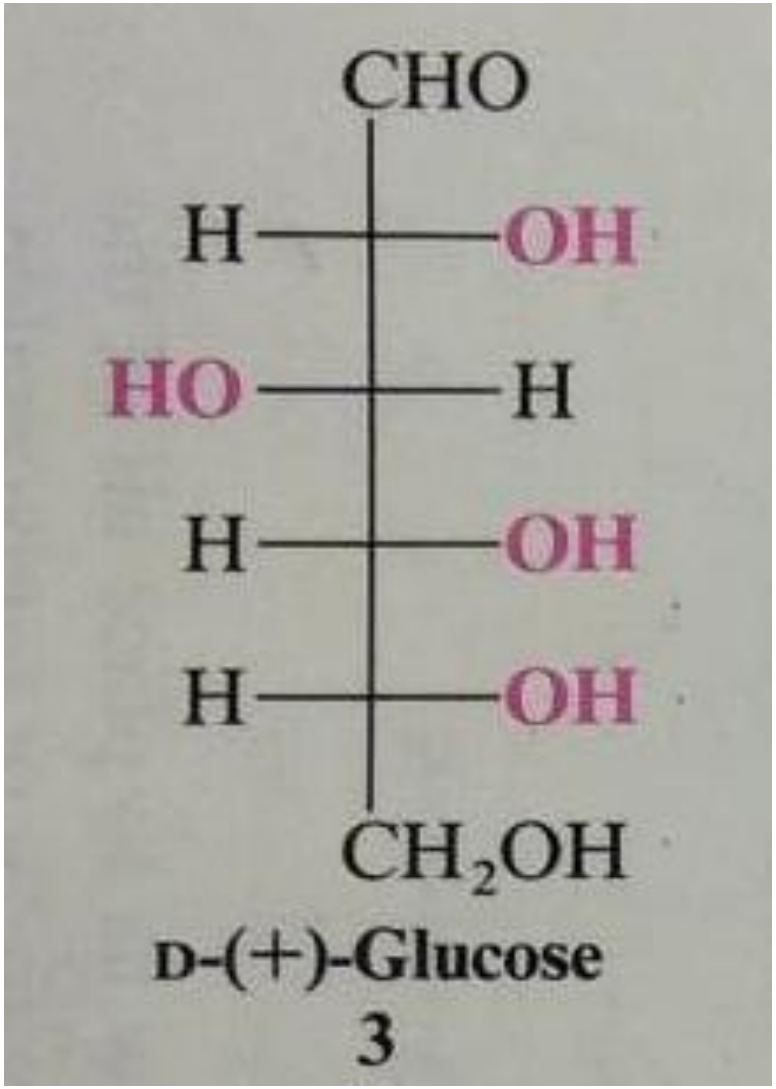


Certain facts were inconsistent with this structure

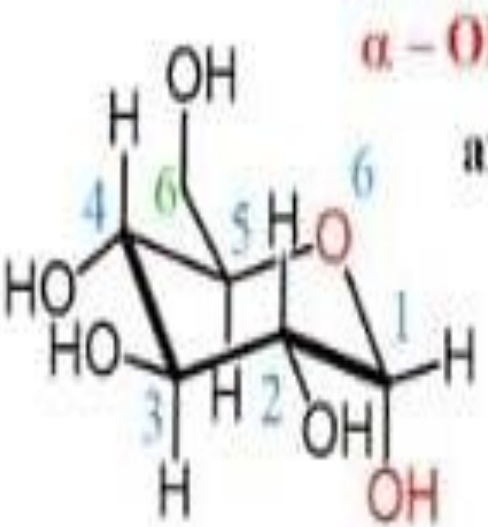
Open structure

Pentahydroxy aldehyde

Conc. very small (less than 0.5%)

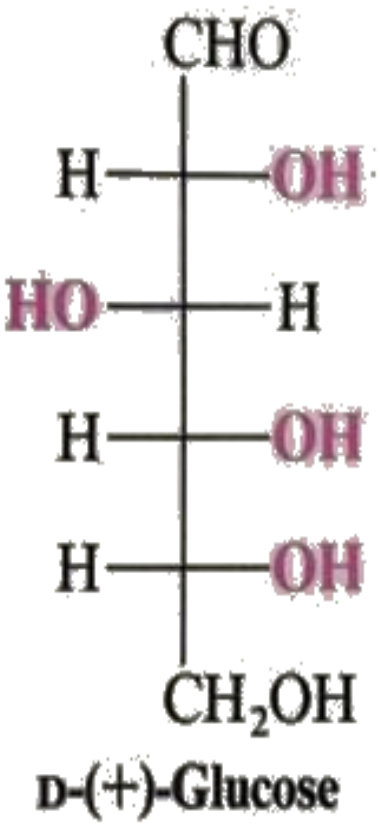
Cyclic structure of (+)-Glucose: cyclic hemiacetal

- **1.** D-(+)- glucose fails to undergo certain reactions typical of aldehydes: **negative Schiff test** and does not form a bisulfite addition product.
- No observable UV or IR absorption band for carbonyl group
- **2.** D-(+)- glucose exists in two isomeric forms which undergo ***mutarotation***

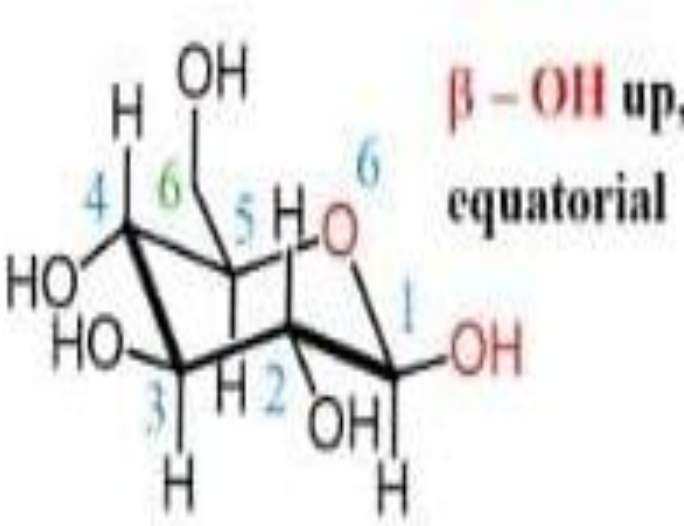


α-D-glucose (36 %)

m.p. 146°C; $[\alpha]_D = +112°$



D-(+)-Glucose



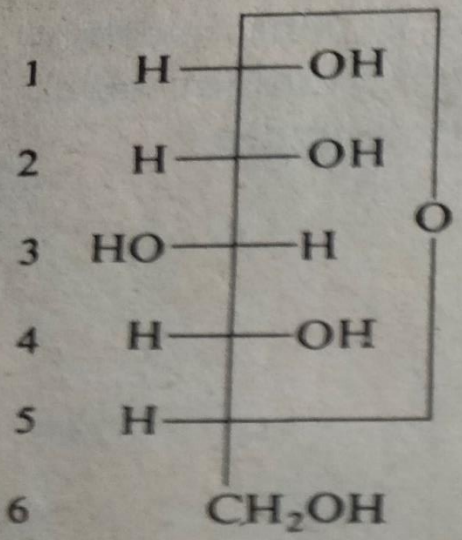
β-D-glucose (64 %)

m.p. 150°C; $[\alpha]_D = +18.7°$

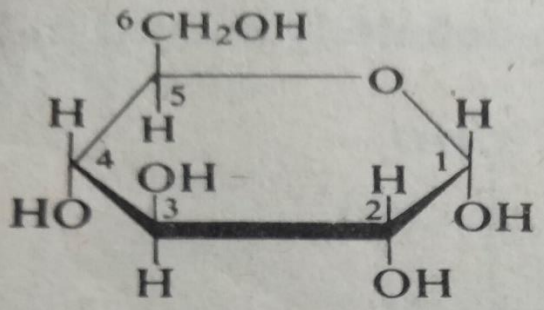
reaches to 52.7° at equilibrium

reaches to 52.7° at equilibrium

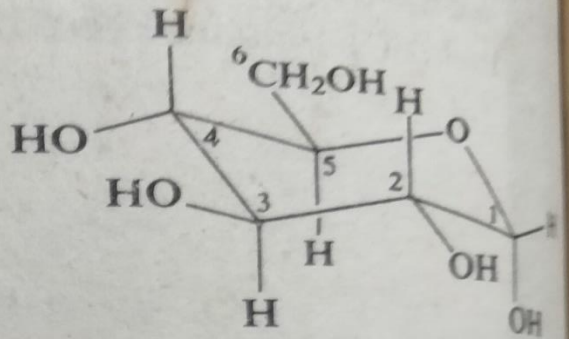
This change in rotation toward an equilibrium value is called ***mutarotation***



IIa

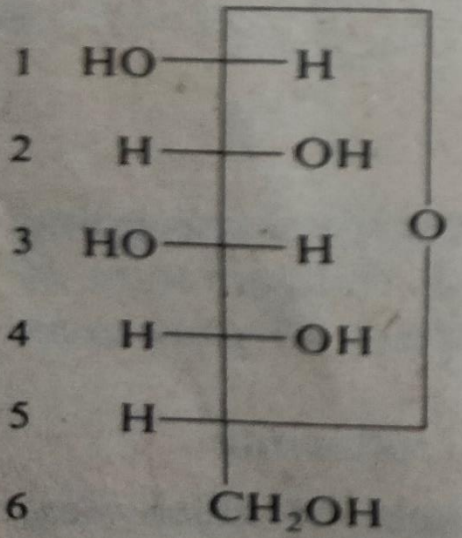


IIb

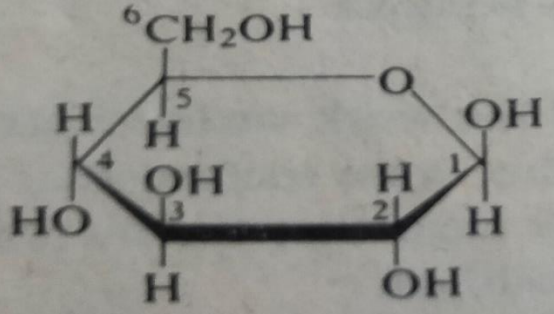


IIc

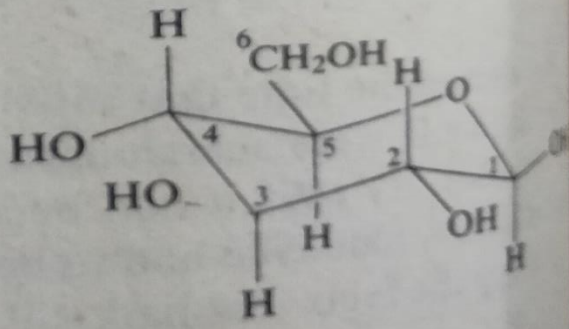
α-D-(+)-Glucose (m.p. 146°, [α] = +112°)



IIIa



IIIb



IIIc

β-D-(+)-Glucose (m.p. 150°, [α] = +19°)

- D-(+)-Glucose is the hemiacetal corresponding to reaction between the aldehyde and C-5 –OH group of the open chain structure. It has a cyclic structure simply because aldehyde and alcohol are part of the same molecule.

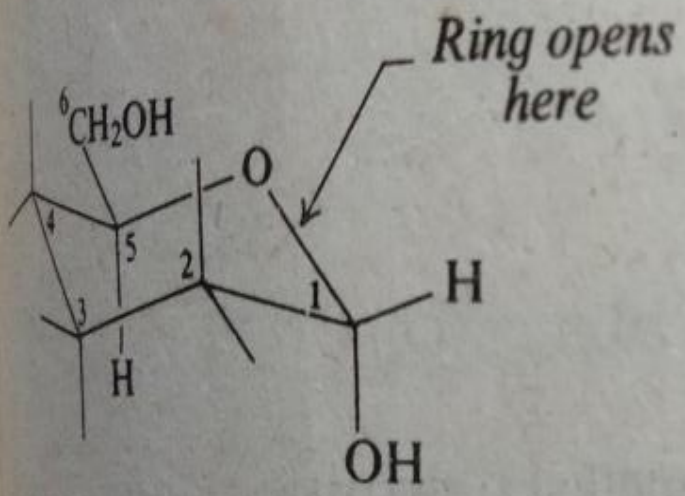
- Two isomeric forms: one more chiral center due to cyclic structure

- α-D-(+)-Glucose and β-D-(+)-Glucose are diastereomers differing in configuration about C-1. Such pair of diastereomers are called **anomers**.

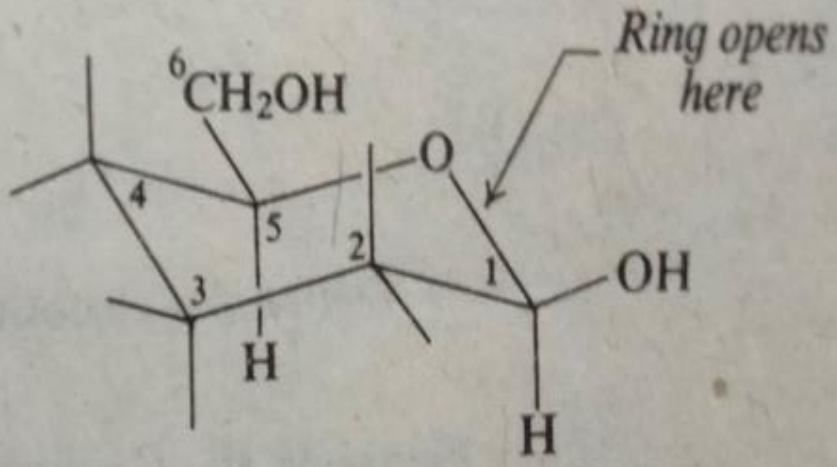
- As hemiacetals, α-D-(+)-Glucose and β-D-(+)-Glucose are readily hydrolyzed by water. In aq. soln. either anomer is converted via open chain form into an equilibrium mixture containing both cyclic isomers. This mutarotation results from the ready opening and closing of the hemiacetal ring.

- The typical aldehyde reactions of D-(+)-Glucose: osazone formation, reduction of Tollens' and Fehling's reagents occurs due to open chain compound; but too low conc. to give other recation, e.g., Schiff test and bisulfite additions

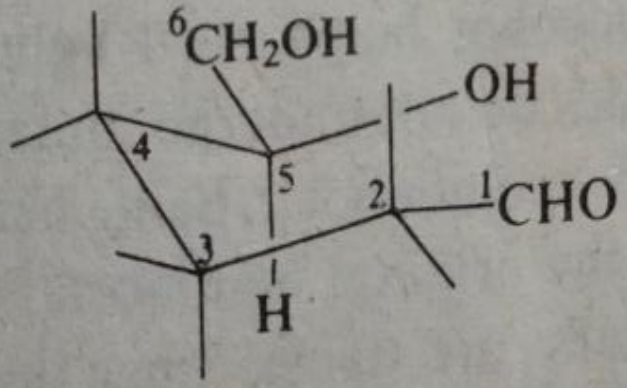
Mutarotation



α-D-Aldohexose



β-D-Aldohexose



Open-chain form

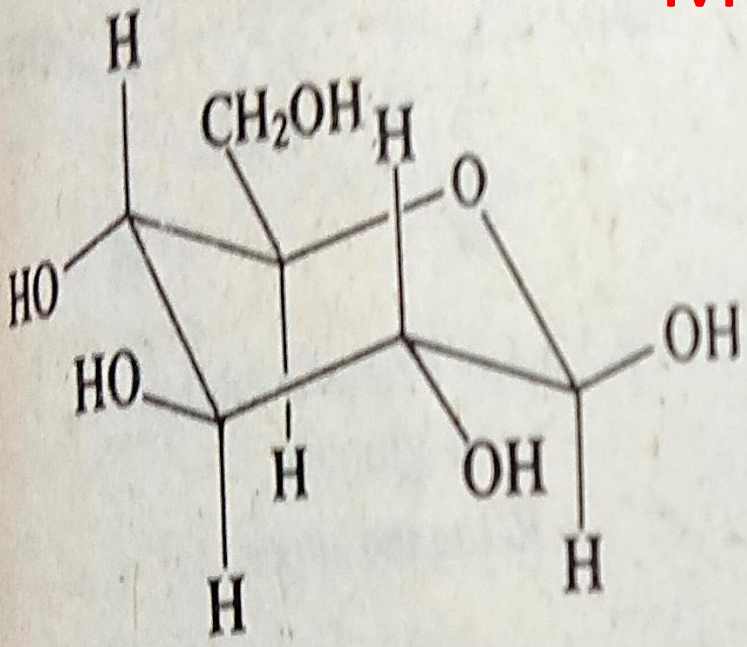
Like D-(+)-Glucose, other mono saccharides exists in anomeric forms capable of mutarotation,

➢**3.** D-(+)- Glucose forms two isomeric methyl D-glucosides. methyl D-glucoside , contains only one $-CH_3$ group but possesses acetal character.

- Like D-(+)-Glucose, other monosaccharides exists in anomeric forms capable of mutarotation, and react with alcohols to yield anomeric glycosides

Acetal formation

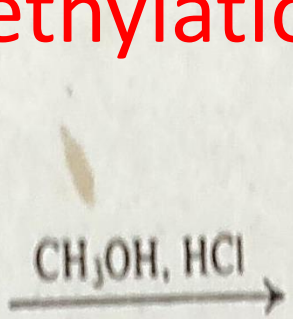
Methylation

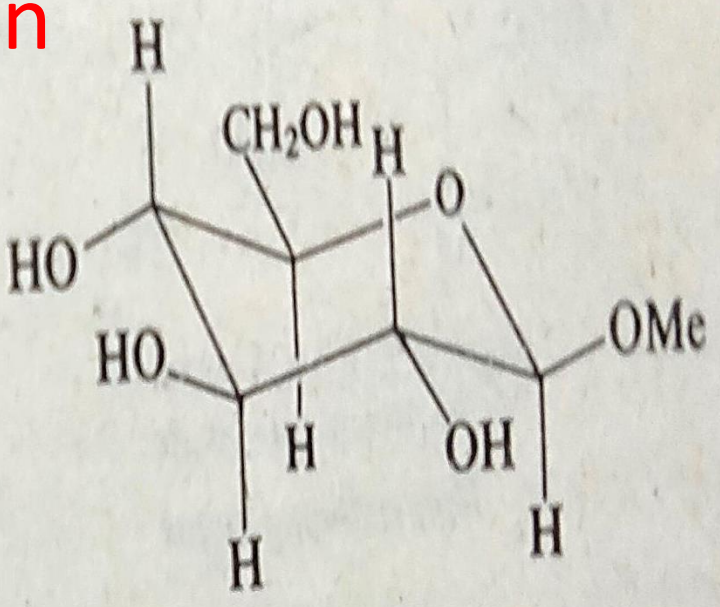


and α-anomer

β-D-(+)-Glucose

Reducing sugar



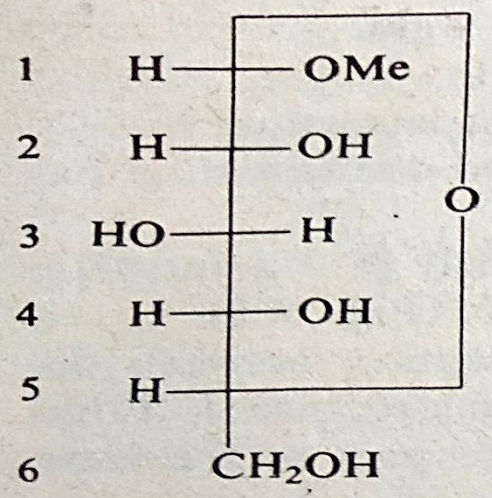


and α-anomer

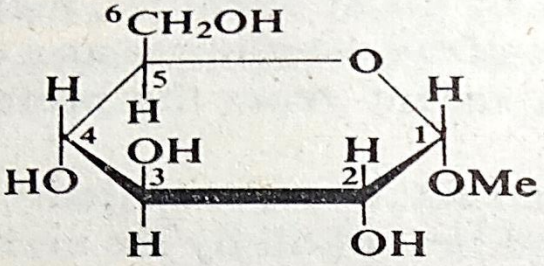
Methyl β-D-glucoside

Non-reducing sugar

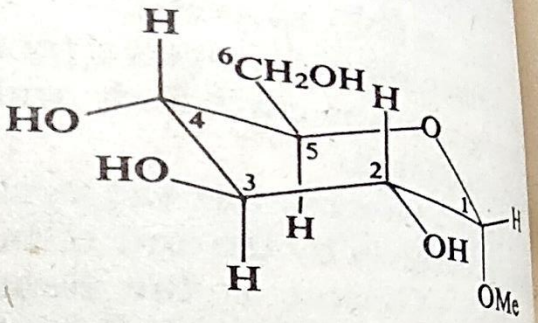
Glucoside anomers: Acetals



IVa



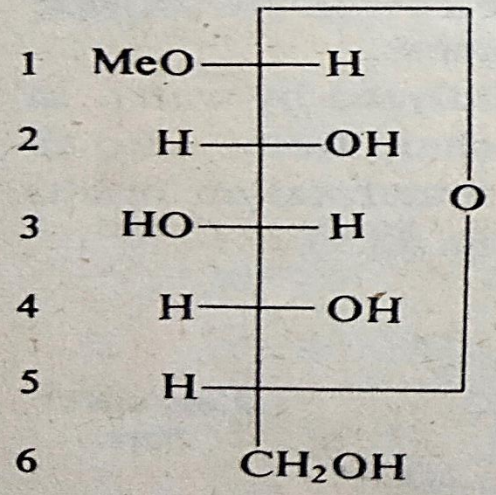
IVb



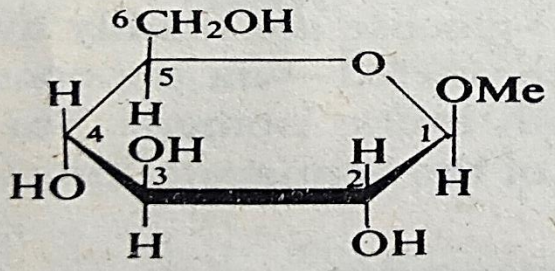
IVc

Methyl α-D-glucoside (m.p. 165°, [α] = +158°)

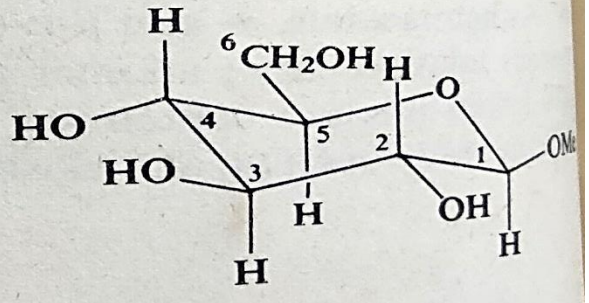
Non-reducing sugars



Va



Vb



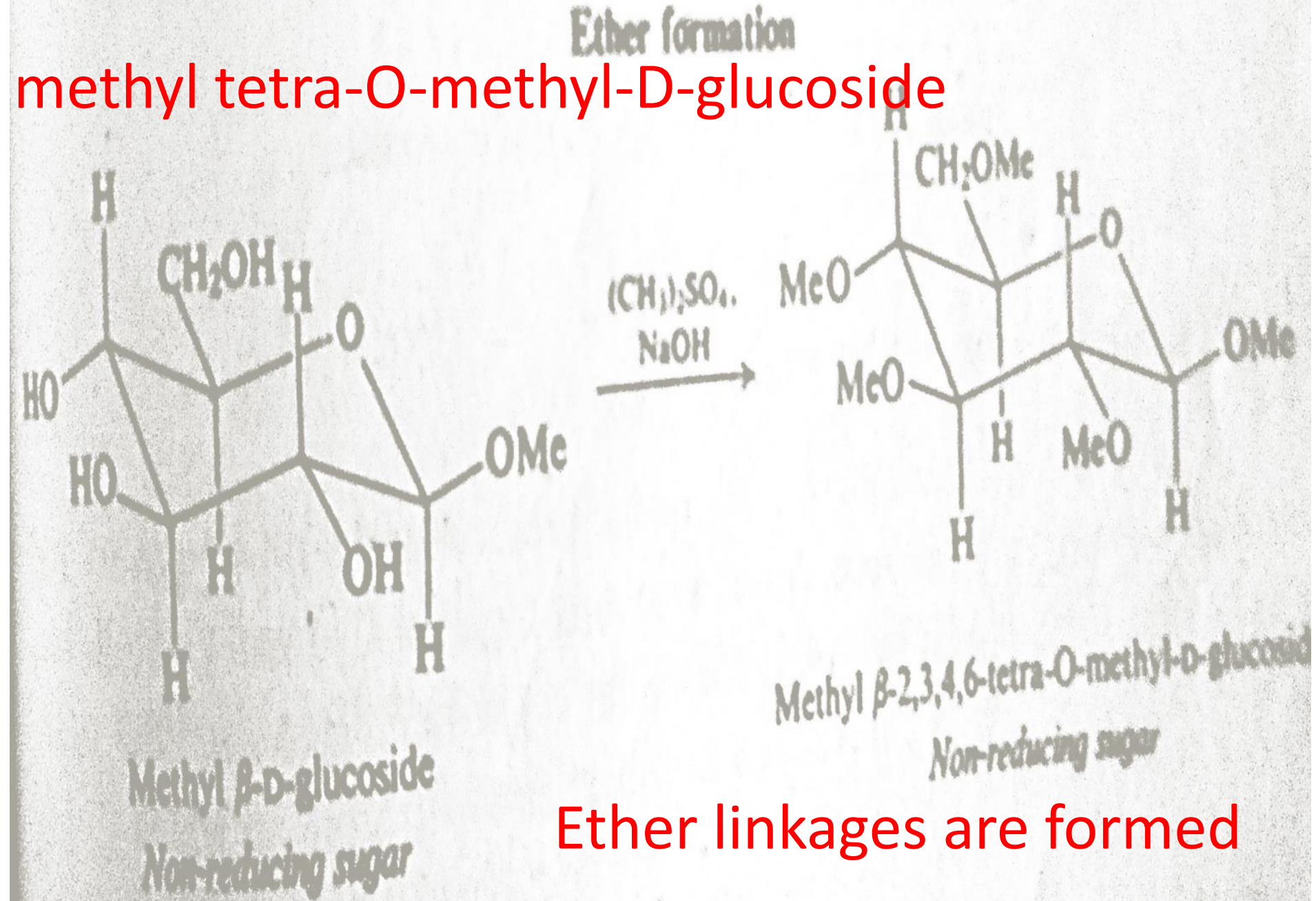
Vc

Do not mutarotate

Methyl β-D-glucoside (m.p. 107°, [α] = −33°)

Figure 28.10. Cyclic structures of methyl D-glucosides.

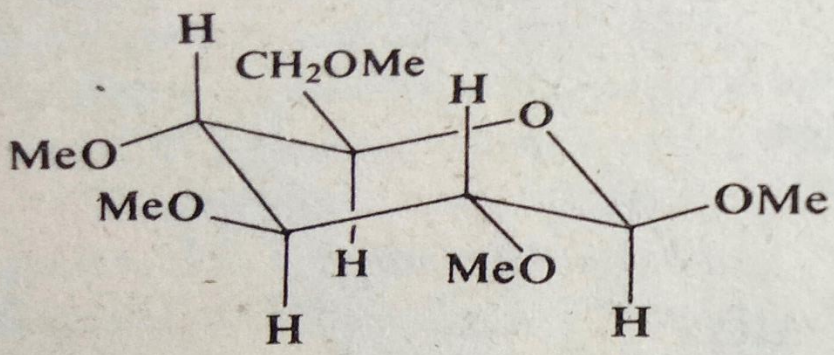
methyl tetra-O-methyl-D-glucoside



Ether linkages are formed

- -OCH_3 attached to every carbon
- Except the one joined to C-1 through the acetal linkage
- **If the six-membered ring is correct , there is an –OCH_3 group on every carbon except C-5.**
- Treatment of the methyl tetra-O-methyl-D-glucoside with dil. HCl removes only one of these –OCH_3, and yields a tetra-O-methyl-D-Glucose.
- Only the reactive acetal linkage is hydrolyzed under these mild conditions, the other four –OCH_3 groups held by ether linkages, remain intact.
- Fully methylated carbohydrate **?**

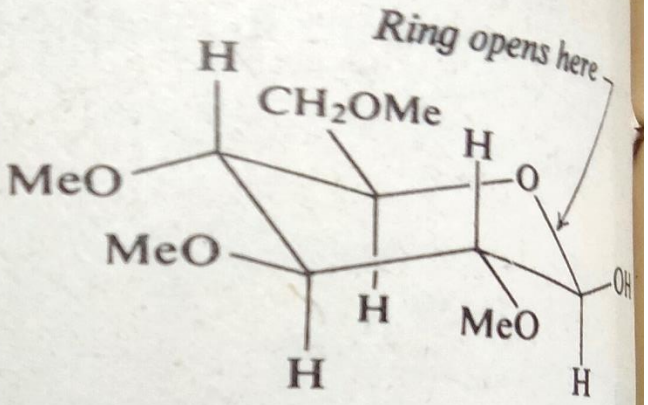
Hydrolysis of an acetal



Methyl β-2,3,4,6-tetra-O-methyl-D-glucoside

Non-reducing sugar

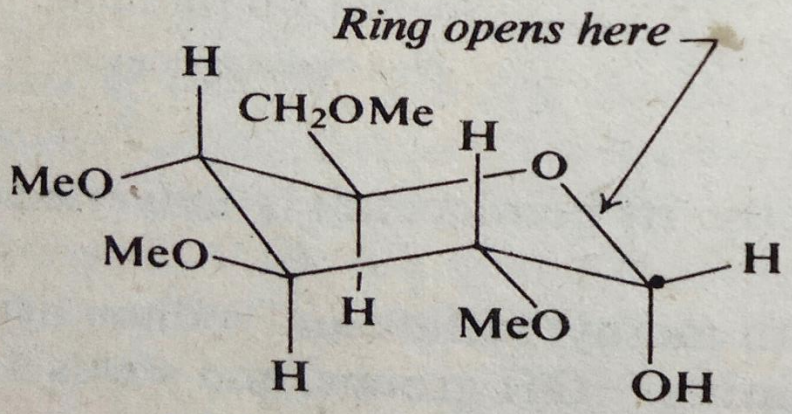
dil. HCl



β-2,3,4,6-Tetra-O-methyl-D-glucose

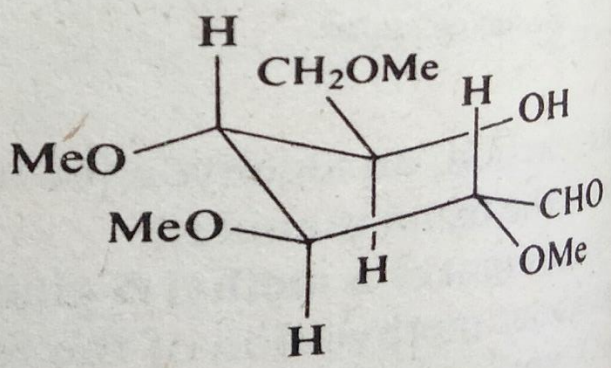
Reducing sugar

tetra-O-methyl-D-glucose



α-2,3,4,6-Tetra-O-methyl-D-glucose

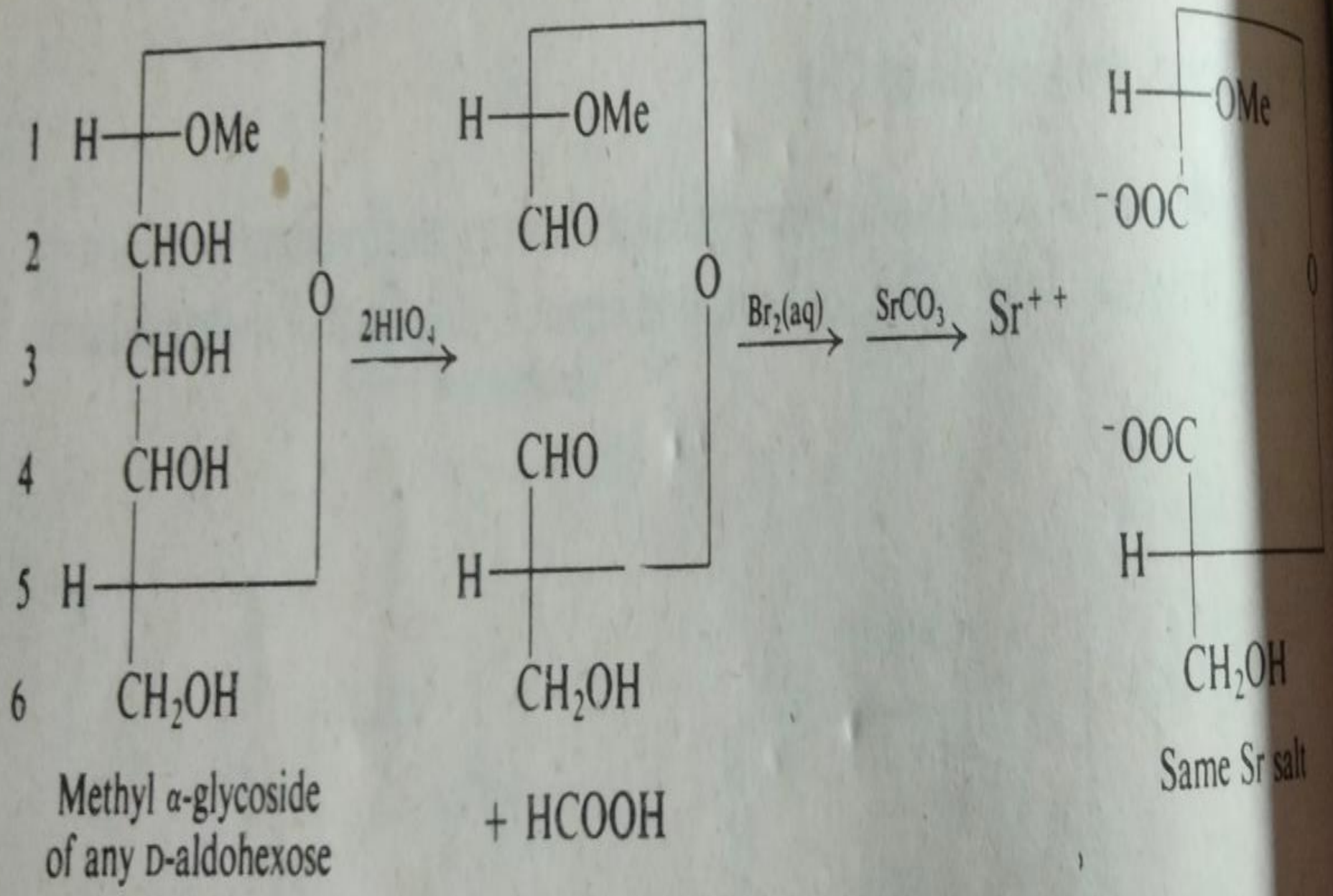
Reducing sugar



- Aldoses and their glycosides have cyclic structures.
- What is Configuration about C-1 in each of these anomeric structures?

- The –OH or –OCH_3 group on C-1 is on the right side in an α-D-anomer and
- On the left in a β-D-anomer for aldoses
- α-D-glucose and methyl α-D-glucoside have same configuration
- β-D-glucose and methyl β-D-glucoside have the same configuration
- Evidence: enzymatic hydrolysis of methyl α-D-glucoside liberates initially the more highly rotating α-D-glucose, hydrolysis of methyl α-D-glucoside liberates initially the more highly rotating α-D-glucose.

- The configuration in C-1 is the same in the methyl α-glucosides of all the D-aldohexoses.
- Evidence: they all yield the same compound upon oxidation by HIO_4.
- Oxidation destroys the chiral centers at C-2, C-3, and C-4, but configuration is preserved about C-1 and C-5. Congratulation about C-5 is the same for all members of the D-family.
- The same products can be obtained from all These glycosides only if they also have the same configuration about C-1.
- C-1-OH is on right in the alpha-D-series
- Left in the beta-D-series.
- Evidence: results of x-ray analysis



Methyl α-glycoside of any D-aldohexose

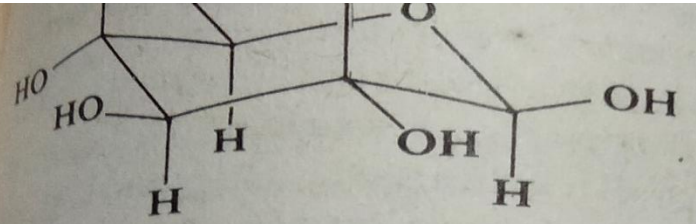
+ HCOOH

Same Sr salt

- Oxidation destroys the chiral centers at C-2, C-3, and C-4, but configuration is preserved about C-1 and C-5. Congratulation about C-5 is the same for all members of the D-family.
- The same products can be obtained from all These glycosides only if they also have the same configuration about C-1.
- C-1-OH is on right in the alpha-D-series
- Left in the beta-D-series.
- Evidence: results of x-ray analysis

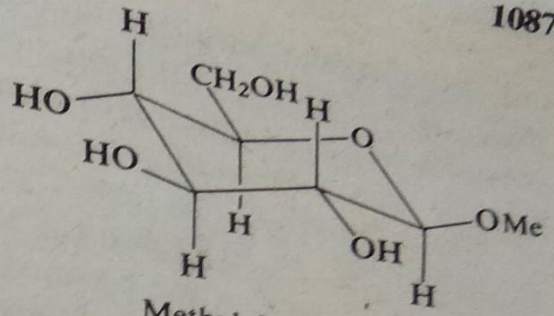
Determination of ring size….

- 5 or/6-membered ?…..
- how to prove?



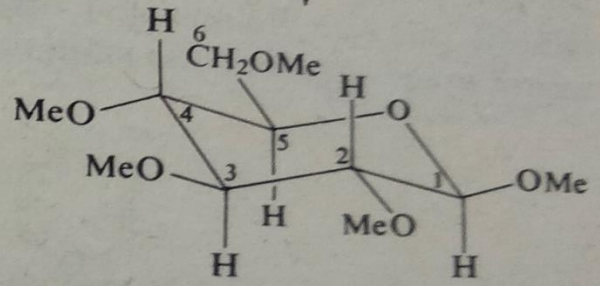
β-D-Glucose

CH_3OH, HCl



Methyl β-D-glucoside

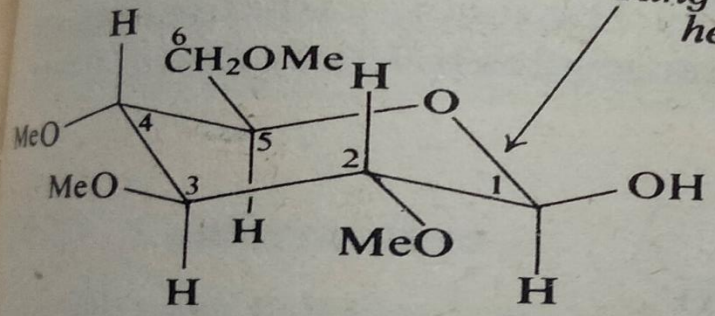
$(CH_3)_2SO_4$, NaOH



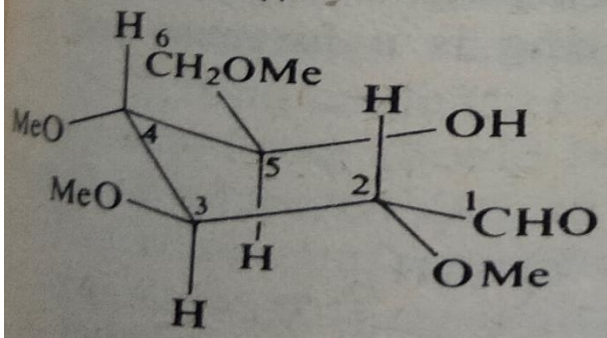
Methyl β-2,3,4,6-tetra-O-methyl-D-glucoside

dil. HCl

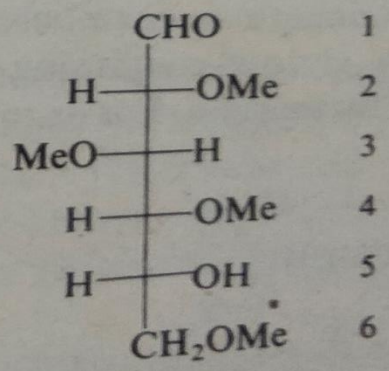
Ring opens here



β-2,3,4,6-Tetra-O-methyl-D-glucose



≡



2,3,4,6-Tetra-O-methyl-D-glucose

Open-chain form

followed by hydrolysis.

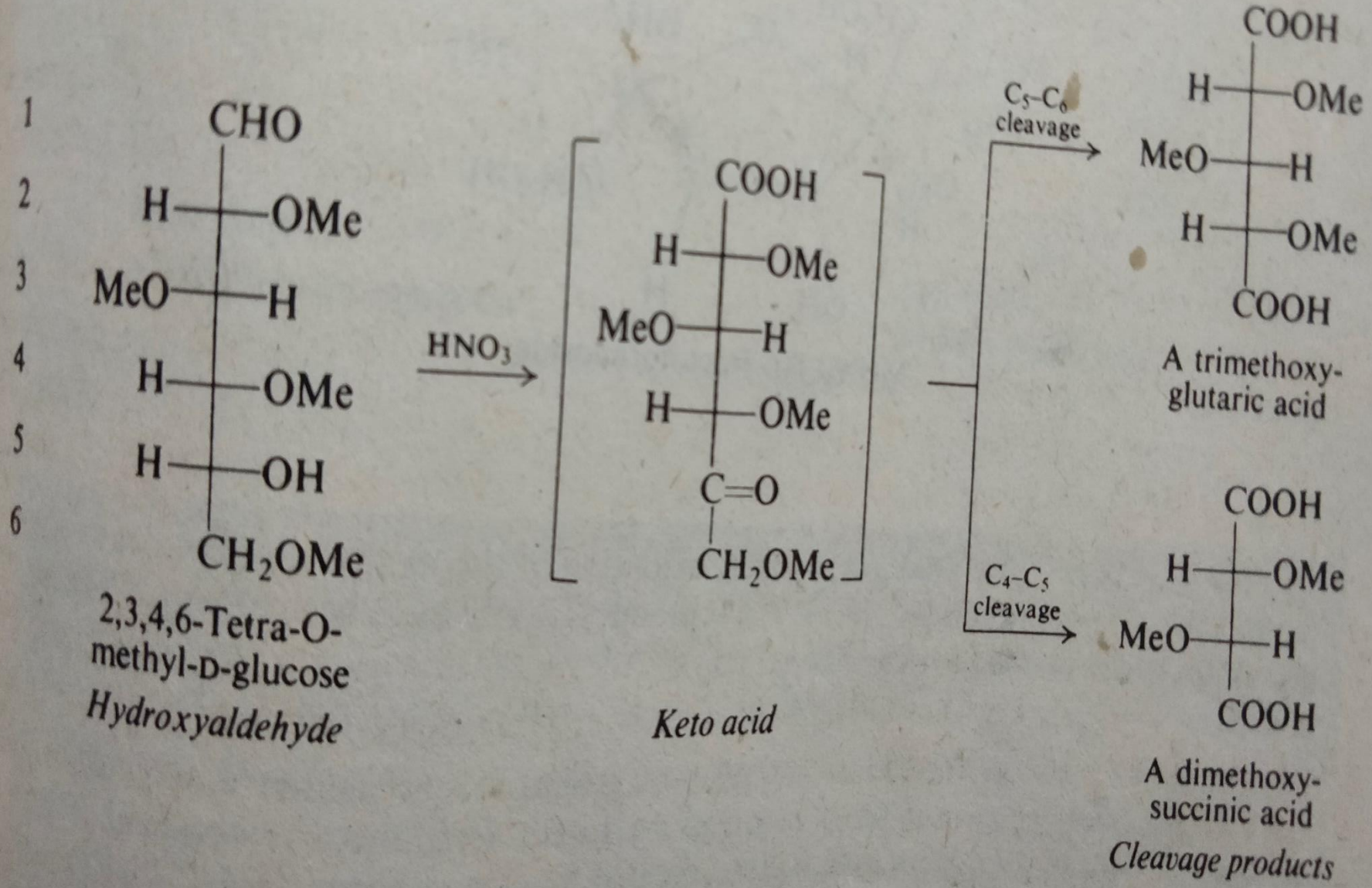


Figure 28.14. Oxidation of 2,3,4,6-tetra-O-methyl-D-glucose.